

off a liability coming due at the maturity of the bond, the investor purchasing the TIPS has eradicated inflation risk. (This is an example of immunization; see chapter 4.) If a retired person can purchase a series of TIPS maturing in different years, then she has locked in a stream of payments that are constant in real terms. This allows her to meet her living expenses in the future, impervious to the effects of inflation. Thus it seems that TIPS are an ideal retirement savings mechanism for retail investors. Zvi Bodie, a professor at Boston University specializing in investment management and pensions, advocates that retail investors should hold close to 100% TIPS in their retirement portfolios.¹⁰

There are several problems with Bodie's advice.

First, purchasing a single TIPS bond provides an inflation hedge for future cash flows immunized at a moment in time—subject to the small indexation lag and the important caveat that the CPI is the right inflation measure for that investor. But perfect cash flow matching assumes that the investor can predict all her future liabilities to immunize them today. Often, she can't really know for certain what liabilities she will face down the road, and some of those retirement expenses are the result of choices about where she works, how long she works, her health, and other considerations along the way.

Second, most investors do not save a lump sum at a moment in time for retirement. Instead, an investor saves a portion of his income at a time over many years. The real yield curve changes across time. As real yields change, the value of the TIPS portfolio also changes, and it changes in such a way such that the TIPS portfolio returns have low correlations with inflation, as I show below. This wouldn't matter if the investor could perfectly forecast his retirement liabilities and he cares only about hedging them using TIPS, but in many situations we care about wealth today as well as wealth at retirement: taking out a mortgage, having a financial cushion to start a business, or even splurging on vacation are all decisions that are impacted by investors' total wealth today.

The third problem with retail investors holding only real bonds for retirement is that at the time of writing in 2013, TIPS are paying negative real yields. If real yields are negative, to secure a payment of \$100 that is immune from inflation risk in five years requires an investor to put away more than \$100 today. Negative real rates mean that TIPS subtract wealth, instead of adding wealth, from investors. In a world of negative real rates, investors investing in risk-free real assets must be prepared to see safe investments decline, not increase, in purchasing power. This critique applies not only to individuals saving for retirement but for all endowments, foundations, and sovereign wealth funds wishing to preserve spending in real terms. While real rates on risk-free bonds may be negative, there are positive expected real risk premiums in other asset classes and strategies (see chapter 7).

A final consideration is that TIPS have sovereign risk. Sovereign risk includes both the risk of a government explicitly defaulting on a real bond—which may

¹⁰ Bodie makes his case in a series of papers and books, most recently in Bodie and Taqqu (2012).